

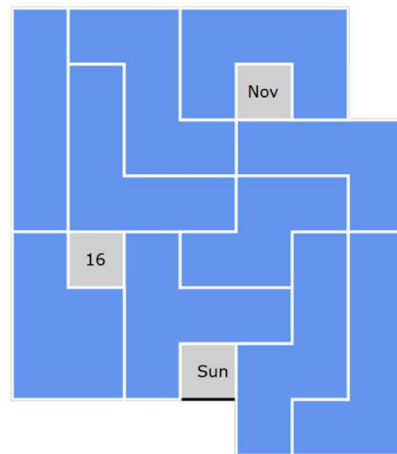
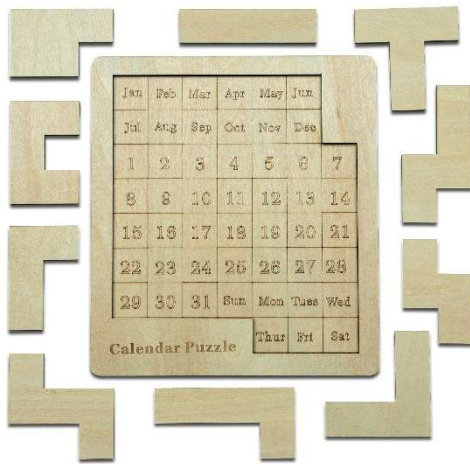
Daily Calendar puzzle solver: (Dan Gilbert style)

Main concept of project:

Creating a GUI integrated program to generate solutions for the classical Dan Gilbert Daily Calendar puzzle

There is more than one valid solution per day and the project focuses on providing atleast one of those valid solutions out of many

The shape of the puzzle and its pieces / an example of a solved puzzle solution



- The puzzle is a 7*8 grid
- The 10 pieces are hard coded
- All rotations and flip orientations of the puzzle are accounted for
- **SFML** is used to display the Calendar solution
- Solving the puzzle will always lead to exactly 3 visible spaces (i.e: month, date, weekday)

CalendarPuzzleSolver.h

Sample output:

```
WELCOME TO THE CALENDAR PUZZLE SOLVER
Calendar puzzle grid:
*-----*
| Jan | Feb | Mar | Apr | May | Jun | X |
*-----*
| Jul | Aug | Sep | Oct | Nov | Dec | X |
*-----*
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
*-----*
| 8 | 9 | 10 | 11 | 12 | 13 | 14 |
*-----*
| 15 | 16 | 17 | 18 | 19 | 20 | 21 |
*-----*
| 22 | 23 | 24 | 25 | 26 | 27 | 28 |
*-----*
| 29 | 30 | 31 | Sun | Mon | Tue | Wed |
*-----*
| X | X | X | X | Thu | Fri | Sat |
*-----*

Enter month (1-12): 12
Enter day (1-31): 21
Enter weekday (1-7) (1:Sun, 2:Mon, ... 7:Sat) : 1

Solving for: 21/12 [weekday 1]...
Solution found! Opening graphical display...
Font loaded successfully from: C:/Windows/Fonts/arial.ttf
|
```

Calendar Puzzle Solution

Solving for: Dec 21, Sun

						X
					Dec	X
						21
			Sun			
X	X	X	X			

Sample output 2:

```
WELCOME TO THE CALENDAR PUZZLE SOLVER
Calendar puzzle grid:
*-----*
| Jan | Feb | Mar | Apr | May | Jun | X |
*-----*
| Jul | Aug | Sep | Oct | Nov | Dec | X |
*-----*
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
*-----*
| 8 | 9 | 10 | 11 | 12 | 13 | 14 |
*-----*
| 15 | 16 | 17 | 18 | 19 | 20 | 21 |
*-----*
| 22 | 23 | 24 | 25 | 26 | 27 | 28 |
*-----*
| 29 | 30 | 31 | Sun | Mon | Tue | Wed |
*-----*
| X | X | X | X | Thu | Fri | Sat |
*-----*

Enter month (1-12): 8
Enter day (1-31): 27
Enter weekday (1-7) (1:Sun, 2:Mon, ... 7:Sat) : 7

Solving for: 27/8 [weekday 7]...
Solution found! Opening graphical display...
Font loaded successfully from: C:/Windows/Fonts/arial.ttf
|
```

Calendar Puzzle Solution

Solving for: Aug 27, Sat

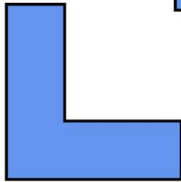
						X
	Aug					X
					27	
X	X	X	X			Sat

Sample output 3 (invalid)

```
WELCOME TO THE CALENDAR PUZZLE SOLVER
Calendar puzzle grid:
*-----*
| Jan | Feb | Mar | Apr | May | Jun | X |
*-----*
| Jul | Aug | Sep | Oct | Nov | Dec | X |
*-----*
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
*-----*
| 8 | 9 | 10 | 11 | 12 | 13 | 14 |
*-----*
| 15 | 16 | 17 | 18 | 19 | 20 | 21 |
*-----*
| 22 | 23 | 24 | 25 | 26 | 27 | 28 |
*-----*
| 29 | 30 | 31 | Sun | Mon | Tue | Wed |
*-----*
| X | X | X | X | Thu | Fri | Sat |
*-----*

Enter month (1-12): 12
Enter day (1-31): 36
Invalid day. Must be 1-31.
```

Note: there is no sample output for “no solution found” here because the puzzle is designed to give solutions for all dates and days.



Enter date in program: Jan 5th, Monday

